

# Data Science

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## Lecture 07: Data Exploration and Preprocessing

Prof. Dr. Michael Granitzer

# Where We Left Off

Three lectures fitted models to tables nobody questioned. Today steps back to KDD steps 1–2 from Lecture 01: what the columns are, whether they can be trusted, and what a plot shows that a summary hides.

## Lecture 01

data is never raw



## Lecture 02

summaries can agree while the data differs  
(Anscombe)

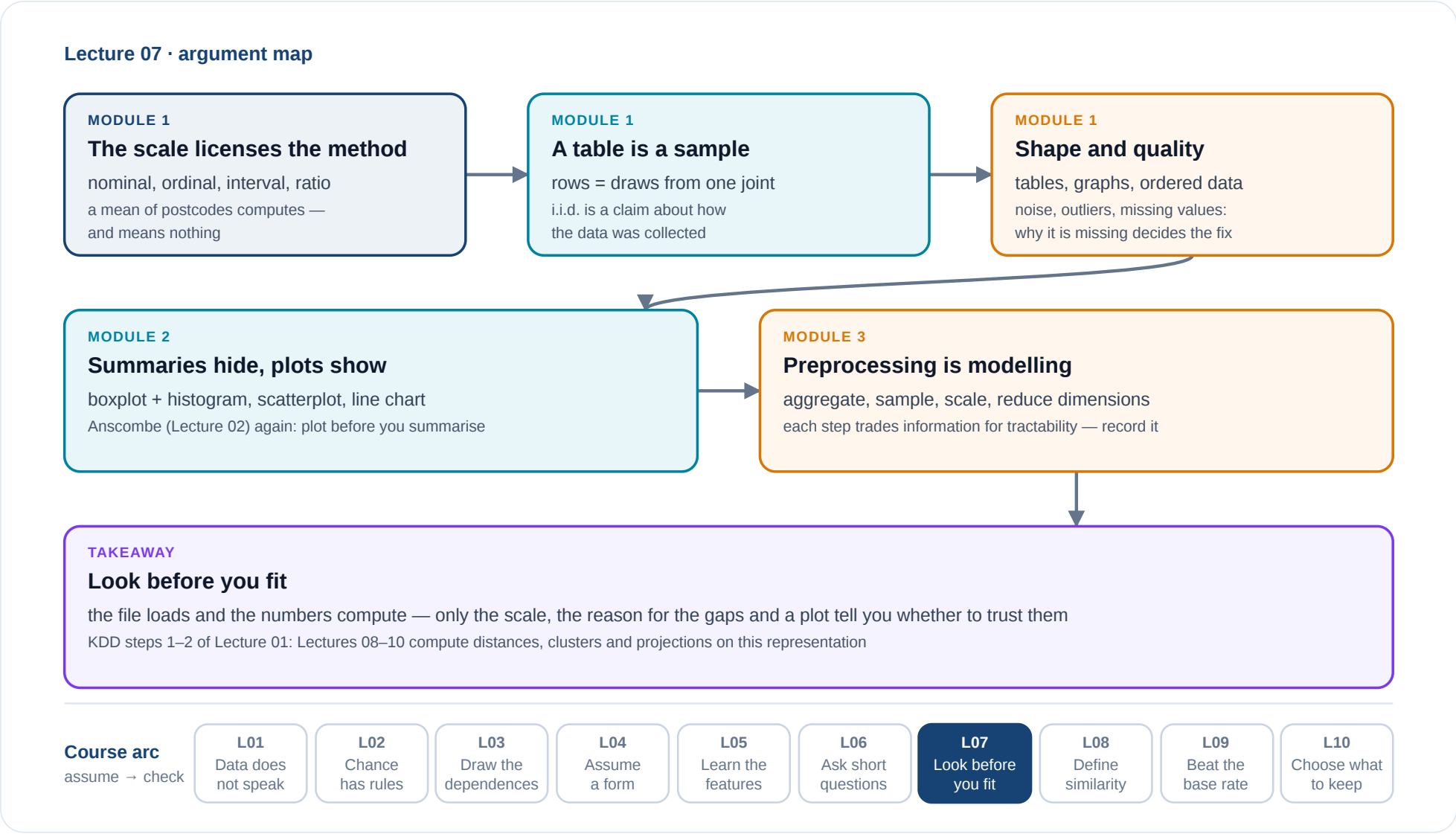


## Today

scale, quality, plots, preprocessing

From Lecture 08 on there is no label to check a model against — looking first stops being good practice and becomes the main check we have.

# Argument Map



# Module 1

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## What Kind of Data Is This?

**Guiding Question:** *The file loads and the numbers compute. Why is that not yet a reason to trust the average?*

# Four Attribute Types

## Nominal

*categorical,  
qualitative*

Labels only: eye  
colour, ZIP code, ID.

Meaningful:  $=$ ,  $\neq$

## Ordinal

*categorical,  
qualitative*

Ordered labels:  
grades, rankings,  
"small/medium/large".

Meaningful:  $<$ ,  $>$

## Interval

*numeric, quantitative*

Differences are  
meaningful, the zero  
point is arbitrary:  
temperature in  $^{\circ}\text{C}$ ,  
calendar dates.

Meaningful:  $+$ ,  $-$

## Ratio

*numeric, quantitative*

A true zero, so ratios  
are meaningful:  
length, mass, counts,  
duration.

Meaningful:  $\times$ ,  $\div$

# Why the Type Decides the Method

## The classic error

Averaging a nominal attribute. The mean of ZIP codes 10115 and 81667 is a number, and it is meaningless.

## The subtle one

20 °C is not "twice as warm" as 10 °C — interval scales have no true zero, so ratios do not transfer.

The attribute type determines which **statistics**, which **distance measures** and which **plots** are permissible. Every method in this course depends on it.

# Mini-Exercise

A customer table records satisfaction (one to five stars) and year of birth. Which scale is each — and whose **mean** would you not report?

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Satisfaction is **ordinal**: its mean assumes equal gaps between the stars — report the median. Year of birth is **interval**: the mean year is fine, but ratios are not ("born twice as late" means nothing); differences such as age are fine.

# Takeaway

An attribute's **scale of measurement** licenses the operations you may perform on it. Choosing a statistic before checking the scale is the most common silent error in data analysis.

# A Table Is a Sample From a Joint Distribution

| customer | payment | satisfaction | born | income (k€) | returned? |
|----------|---------|--------------|------|-------------|-----------|
| 1        | card    | 4            | 1987 | 52          | yes       |
| 2        | invoice | 2            | 1995 | 31          | no        |
| 3        | card    | 5            | 1979 | 78          | no        |
| ...      | ...     | ...          | ...  | ...         | ...       |

## Each column is a random variable

$X_{\text{payment}}, X_{\text{satisf.}}, \dots$  — a function from customers to values, exactly as a random variable was defined.

## Each row is one draw

The whole table is  $n$  draws from one joint distribution  $p(x_1, \dots, x_d)$  that nobody has written down.

# The Scale Suggests the Model

| attribute    | scale                   | a first model                                              |
|--------------|-------------------------|------------------------------------------------------------|
| returned?    | nominal, two values     | <b>Bernoulli</b> ( $p$ ) — one parameter                   |
| payment      | nominal, $k$ values     | <b>categorical</b> ( $p_1, \dots, p_k$ ), $\sum_j p_j = 1$ |
| satisfaction | ordinal                 | categorical — the order is not used, or an ordinal model   |
| born         | interval                | normal, on a restricted range                              |
| income       | ratio, positive, skewed | log-normal, or normal on log income                        |

The scale limits the model as it limits the statistics: a normal distribution for ZIP codes is as meaningless as their mean.

# What "i.i.d." Claims

The standard reading of a table rests on one assumption: the rows are **independent and identically distributed**.

## Independent

One customer's row tells us nothing about another's.

## Identically distributed

Every row comes from the **same** joint distribution.

Which of these datasets break it: daily sales of one shop · two visits by the same customer · pixels of one image · a survey sent to a random sample?

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Only the random survey is plausibly i.i.d. Time series, repeated measurements and spatial data all violate it — the arrangement *is* information.

# Takeaway

A dataset is  $n$  **draws from an unknown joint distribution**: columns are random variables, rows are draws, and the scale of each column suggests its model. Reading a table this way rests on an assumption — **i.i.d.** — that should be checked, not presumed.

# Four Shapes of Data

## Record data

Objects × attributes  
— the familiar table.  
Rows are *treated as* independent.

## Graph data

Objects carry **links** to other objects: web pages, social networks, citations.

## Fields / grids

Objects sit in a **spatial** arrangement: images, sensor lattices, simulations.

## Spatio-temporal

Position **and** time: monthly land temperatures, traffic movement, trajectories.

The shape decides which methods even apply. A table whose rows are treated as independent can be shuffled without loss; a grid or a time series cannot — the arrangement *is* information. Independence of rows is an assumption, not a property of the file.

# Three Data Quality Problems

## Noise

Random distortion of measured values. Cannot be removed, only modelled or smoothed.

## Outliers

Values far from the rest.  
Either an **error** to remove or the **signal** you were looking for — fraud, faults, discoveries.

## Missing values

Absent entries. Handling depends on **why** they are missing, which the data itself does not tell you.

"Outlier" is not a property of a data point; it is a judgement about a model of what is normal. Deleting outliers deletes evidence against that model.

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No. Deletion removes the coldest days, so every statistic afterwards — the mean temperature first — is biased upwards. Had the gaps been random transmission dropouts, deletion would be roughly harmless. The file looks the same in both cases; only the **mechanism** differs.

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The shape of the data decides which methods apply; the **reason** for a quality problem decides how it may be handled. Neither is visible in the file.

# The Churn Shop's Customer Table

| customer | contract | satisfaction | tenure (months) | postcode | churned? |
|----------|----------|--------------|-----------------|----------|----------|
| 1        | monthly  | 4            | 3               | 94032    | yes      |
| 2        | yearly   | —            | 27              | 80331    | yes      |
| 3        | yearly   | 5            | 41              | 94032    | no       |
| ...      | ...      | ...          | ...             | ...      | ...      |

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### SCALES

Contract, postcode, churned: nominal.  
Satisfaction: ordinal. Tenure: ratio. The mean postcode computes — and means nothing.

### WHY IS SATISFACTION MISSING?

Customers who left stop answering: 40 of 100 churners gave no rating, 45 of 900 stayers. Drop those rows, and churn falls from **10%** to **6.6%**.

# Module 2

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## Seeing the Data

**Guiding Question:** *Summary statistics answer the questions you thought to ask. What shows you the ones you did not?*

# Visualization Is Everywhere

Newspaper tables, subway maps, weather charts, stock graphs, a 3-D reconstruction of an injured knee, demographic bar charts, social-network layouts.

**What do these have in common?** All of them present data in a form the eye processes faster than the equivalent numbers.

# Why It Works

## Bandwidth

Vision is the human sense with the highest data rate — pattern detection runs in parallel and pre-attentively.

## Interaction

A picture is a natural surface for exploring: zoom, filter, select, compare.

## Discovery

Visualization surfaces the **unexpected** — the structure nobody thought to compute a statistic for.

Summary statistics answer questions you already asked. A plot lets the data ask you one — which is exactly what Anscombe's quartet demonstrated.

Visualization is not decoration for results; it is an **instrument of analysis**, and the only one that can show you what you did not think to measure.

# The Boxplot

A visual summary of a distribution built from quartiles:

## The box

Spans  $Q_1$  to  $Q_3$  — the interquartile range, the middle half of the data.

## The line

The **median** inside the box. Its position shows skew.

## The whiskers

(Tukey) Reach to the most extreme data points inside  $[Q_1 - 1.5 \cdot \text{IQR}, Q_3 + 1.5 \cdot \text{IQR}]$ . Points outside are drawn singly as **outliers**.

The  $1.5 \times \text{IQR}$  rule is a **convention**, not a test. It flags candidates for inspection, not errors for deletion.

# The Histogram

## What it shows

The **shape** of a distribution: modality, skew, gaps and clusters that a boxplot hides entirely.

## What it hides

Everything depends on the **bin width**. Too wide and structure vanishes; too narrow and noise looks like structure.

A bimodal distribution and a unimodal one can produce the same boxplot. Always look at one histogram before trusting a five-number summary.

# Mini-Exercise

Two datasets have the same minimum,  $Q_1$ , median,  $Q_3$  and maximum — so the same boxplot.  
Must their histograms look alike?

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No. The five numbers fix five points; the mass between them can be arranged freely — spread evenly inside the box, or piled into two clusters at its edges. Same quartiles, opposite stories: Anscombe's quartet again.

# Mini-Exercise

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No. The five numbers fix five points; the mass between them can be arranged freely — spread evenly inside the box, or piled into two clusters at its edges. Same quartiles, opposite stories: Anscombe's quartet again.

A boxplot compares distributions compactly; a histogram shows their shape. **Neither alone is sufficient**, and the boxplot's outlier rule is a convention you may overrule.

# The Scatterplot

An X–Y plot of two dimensions against each other — the standard instrument for inspecting a **relationship** rather than a distribution.

With more than two dimensions there are four options:

## Subsetting

Let the user choose which dimensions to display.

## Reduction

Project to fewer dimensions (see preprocessing, and projections later).

## Embedding

Map extra dimensions to other channels: colour, size, shape.

## Multiple displays

Superimpose, or juxtapose in a **scatterplot matrix**.

# What a Scatterplot Reveals

Linear association, non-linear structure, clusters, outliers and heteroscedasticity — most of which no single correlation coefficient reports.

**Pearson correlation**  $r = \frac{\text{cov}(X, Y)}{s_X s_Y} \in [-1, 1]$ : the covariance measured in units of the two standard deviations. It measures **linear** association only.

A correlation of 0 is consistent with a perfect parabola. The scatterplot shows it; the coefficient does not.

# Tufte's Principle

**Maximise the data-ink ratio:** every drop of ink on the page should carry information. Delete anything that does not — heavy gridlines, boxes, redundant tick marks, decorative fills.

**Range frames** apply the principle to the axes: draw each axis only over the interval the data actually spans, so the frame itself reports the minimum and maximum.

The exercise: take a finished chart and ask of each element, *what would I lose by deleting this?*

A scatterplot shows the structure a coefficient summarises away. Beyond two dimensions you must **subset, reduce, embed or juxtapose** — and every mark you draw should earn its ink.

# When a Line Is Justified

Connecting points with a line asserts an **order** between them — and often an interpolation.

**Data requirement:** one or more quantitative value attributes on a comparable scale, plus one **ordered** key attribute (usually time).

Connecting nominal categories with a line claims an order that does not exist. The line is a statement, not a decoration.

# Aspect Ratio Is Not Cosmetic

Readers compare trends by the **orientation** of the line segments, so the shape of the plotting area changes what they conclude. The same series looks flat in a wide frame and dramatic in a tall one.

**Banking to 45°** chooses the aspect ratio so that the segments' average absolute orientation is 45°: differences in orientation are judged best near  $\pm 45^\circ$  (Cleveland et al. 1988).

## Additional channels

Colour and line type distinguish series — but only for a handful of them.

# Multiple Series

## Overlaid

Best for comparing a few series directly against each other.

## Stacked

Each series is drawn on top of the previous one. Shows the total well — and makes individual series **hard to read**, since each sits on a moving baseline.

## Sorted / small multiples

One panel per series, ordered by some property. Scales to many series where overlaying fails.

A line asserts order and interpolation; the **aspect ratio** decides what the reader sees and stacking trades the readability of the parts for the readability of the whole.

# Module 3

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## Preprocessing

**Guiding Question:** *Lecture 01 claimed that preprocessing is modelling. Which choices made here change what a model can learn?*

# Aggregation

Combine several objects or attributes into one — customer purchases summarised by store, by month, by price band.

## Gains

Fewer objects to process; **less variation**, hence more stable estimates; the natural operation of a data warehouse.

## Costs

Detail is gone, irreversibly. Effects that exist only within a group become invisible.

# Sampling

Work with a subset of the objects: random sampling with or without replacement, or **stratified** sampling that preserves group proportions.

Sampling always loses information. The question is whether it loses anything that matters — a sufficiently large sample retains the overall structure.

Sample a sine wave once per period and every sample has the same value: the reconstruction is a flat line — not merely rougher, but **wrong**.

# Scaling and Encoding

## Scaling numeric attributes

Min–max:  $x' = \frac{x - \min}{\max - \min} \in [0, 1]$

Standardisation:  $z = \frac{x - \bar{x}}{s}$  — mean 0, standard deviation 1

## Encoding categorical attributes

**Nominal** → one-hot: one 0/1 column per value, so no false order.

**Ordinal** → ranks 1, 2, 3, ..., which keep the order.

Two customers differ by 1,000 € in income and 30 years in age. Unscaled, their distance is  $\approx 1,000$ : age is invisible. Standardised ( $s = 10,000$  € and 15 years) the differences are 0.1 and 2 — now age dominates.

# Dimensionality Reduction

**The curse of dimensionality.** As the number of attributes grows, the data becomes sparse: distances between points concentrate, neighbourhoods lose meaning, and the volume to be covered grows exponentially.

## The problem

Many features are irrelevant, noisy or correlated.

## The cost

More computation, more parameters, more variance — and weaker distance-based methods.

## The response

**Feature selection** keeps a subset; **feature extraction** builds new combined dimensions.

Every preprocessing step **trades information for tractability**. The trade is usually worth it — but it is a modelling decision, and it should be recorded as one.

## Image Credits & References

Cleveland, William S. and McGill, Marylyn E. and McGill, Robert (1988). The Shape Parameter of a Two-Variable Graph. *Journal of the American Statistical Association*.